

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R,S) / S6 (PT) (R,S) Examination May 2024 (2019 Scheme)

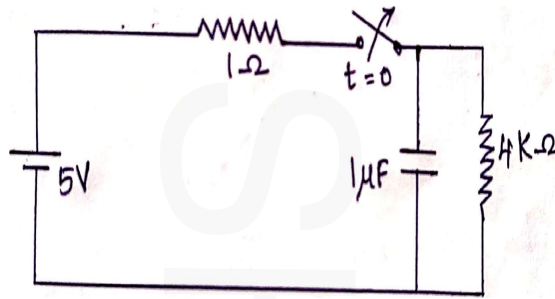
Course Code: EET 308**Course name: COMPREHENSIVE COURSE WORK**

Max. Marks: 50

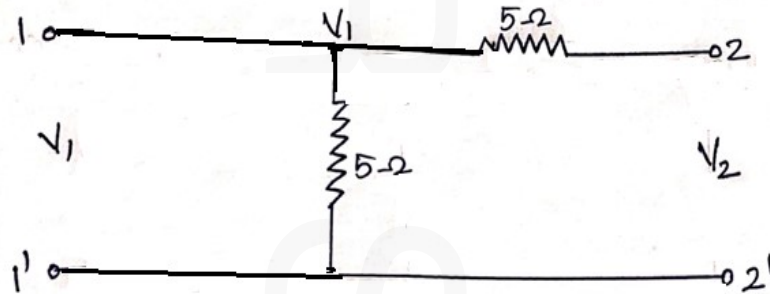
Duration: 1 Hour

- Instructions:**
- (1) Each question carries one mark. No negative marks for wrong answers
 - (2) Total number of questions: 50
 - (3) All questions are to be answered. Each question will be followed by 4 possible answers of which only ONE is correct.
 - (4) If more than one option is chosen, it will not be considered for valuation.

1. In the analysis of a transistor circuit we usually use
 - a) Norton's theorem
 - b) Thevenin's theorem
 - c) Reciprocity theorem
 - d) Superposition theorem
2. A 5A source has a shunt internal resistance of 100 ohm. The maximum power that can be delivered to a load is
 - a) 625W
 - b) 1250W
 - c) 325W
 - d) 900W
3. Reciprocity theorem cannot be applied to the circuits having
 - a) Bilateral elements
 - b) Non-linear elements and multi-sources
 - c) Linear elements
 - d) Only one independent sources
4. A two-port network is said to be reciprocal if and only if
 - a) $Z_{12}=Z_{21}$
 - b) $Z_{12}=Z_{22}$
 - c) $Z_{11}=Z_{12}$
 - d) $Z_{11}=Z_{22}$
5. If a capacitor is energised by a symmetrical square wave current, then the steady state voltage across the capacitor will be
 - a) Step function
 - b) Square wave
 - c) Impulse function
 - d) Triangular wave
6. In RLC circuits, the current at resonance is
 - a) Maximum in parallel resonance and minimum in series resonance
 - b) Maximum in series resonance and minimum in parallel resonance
 - c) Maximum in both series and parallel resonance
 - d) Minimum in both parallel resonance and series resonance
7. The switch in the circuit has been closed for a long time. It is opened at $t=0$. At $t=0^+$, the current through the $1\mu\text{F}$ capacitor is



- a) 0A b) 1.25A c) 1A d) 5A
8. For a RL circuit, $R=3.14\ \Omega$ and $L=1\text{H}$ connected across 220 V, 50Hz supply the quality factor is
- a) 1 b) 50.001 c) 314.16 d) 100
9. The value of the parameter A for two-port network shown in figure is

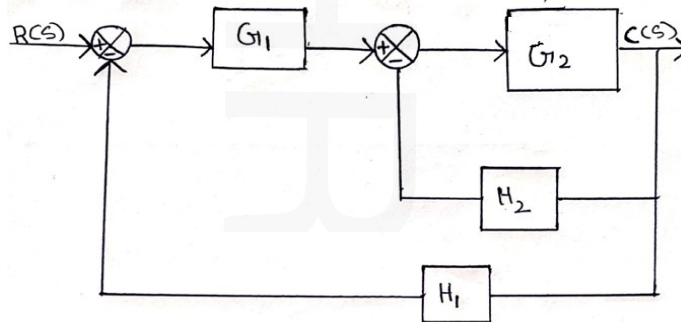


- a) Zero b) 0.5 c) 2 d) 1
10. If an ideal voltage source and ideal current source are connected in series, the combination
- a) Has the same properties as a current source alone b) Has the same properties as a voltage source alone c) Has the same properties as the source which has a higher value d) Results in the branch being redundant
- 11 The hexadecimal number $(B6C7)_{16}$ is equivalent to decimal number
- a) 49761 b) 46791 c) 47691 d) 49167
- 12 Which logic gate is similar to the function of two series switches
- a) AND b) OR c) NAND d) All of the above
- 13 The Boolean function $A+BC$ is a reduced form of
- a) $AB+BC$ b) $(A+B)(A+C)$ c) $A'B+AB'C$ d) $(A+C)B$
- 14 A basic S-R flip flop can be constructed by cross coupling of which basic logic gates
- a) AND or OR gates b) XOR and XNOR gates c) NOR or NAND gates d) AND or NOR gates
- 15 The present outputs of the flip-flops of a 3-bit synchronous down counter are 110. After 7 clock pulses the output changes to
- a) 110 b) 111 c) 101 d) 100
- 16 The ADC having highest conversion speed is

- a) Dual slope ADC b) Successive approximation ADC c) Flash type ADC d) Servo type ADC
- 17 The sequential circuit in which the output depends only on the present state of flip-flops is called a
- a) Mealy circuit b) Moore circuit c) Transition circuit d) None of the above
- 18 Which of the following circuit can be used as parallel to serial converter.
- a) Multiplexer b) Demultiplexer c) Decoder d) Encoder
- 19 The decimal number -19 is expressed in 2's complement form as
- a) 10010011 b) 11101101 c) 11011100 d) 00110011
- 20 A shift register in which the output of the last flip-flop is connected to the input of the first flip-flop
- a) BCD counter b) Parallel counter c) Ripple counter d) Ring counter
- 21 In a simplex wave winding, the commutator pitch is approximately equal to
- a) Pole pitch b) Twice the pole pitch c) Thrice the pole pitch d) Twice the front pitch
- 22 A 6 pole lap wound generator has 300 conductors, the emf induced per conductor being 5 V. The generated voltage of the generator is
- a) 60V b) 1500V c) 360V d) 250V
- 23 A compensating winding is an auxiliary winding embedded in slots in
- a) the armature b) the pole faces of main poles c) partly in pole faces, partly in armature d) the interpoles
- 24 A DC motor which can provide zero speed regulation at full load without any controller is
- a) Series b) Shunt c) Cumulative compound d) Differential compound
- 25 If the back emf in a DC motor vanishes suddenly, the motor will
- a) Run at very high speed b) Run at very slow speed c) Have armature winding burnt d) Not run at all
- 26 When an electric train is moving down a hill, the DC motor will operate as
- a) DC series motor b) DC shunt motor c) DC series generator d) DC shunt generator
- 27 The full load copper loss of a transformer is 1600 watts. At half-load the copper loss will be
- a) 80 W b) 400W c) 1600W d) 6400W
- 28 A single-phase transformer has turns ratio of 10. The primary winding has resistance of 2.2 ohms and secondary has resistance of 0.03 ohms. The total resistance as referred to primary is
- a) 2.5Ω b) 5.2Ω c) 2.23Ω d) 2.203Ω
- 29 The purpose of performing Sumpner's test is mainly to find out
- a) Regulation b) Efficiency c) Temperature rise on full load d) None of the above
- 30 Delta/Star transformers work satisfactorily when
- a) Load is unbalanced only b) Load is balanced only c) On balanced as well as unbalanced loads d) None of the above
- 31 A generating station has a maximum demand of 20 MW and a connected load of 40 MW. The generated units are 4.36×10^7 per annum. The demand factor is

- a) 2 b) 0.5 c) 2/3 d) 1/3
- 32 Skin effect results in
- a) Reduced effective resistance but increased effective internal reactance of the conductor b) Reduced effective resistance as well as effective internal reactance c) Increased effective resistance but reduced effective internal reactance of the conductor d) Increased effective resistance as well as effective internal reactance
- 33 Corona loss will increase with
- a) Increase in conductor size and decrease in supply frequency b) Decrease in conductor size and decrease in supply frequency c) Decrease in conductor size and increase in supply frequency d) Increase in conductor size and increase in supply frequency
- 34 Calculate the PSM of a 6A overcurrent relay having a current setting of 150% connected to a supply circuit through 420/6A CT when the circuit carries a fault current of
- a) 6.67 b) 4.6 c) 8 d) 8.32
- 35 Relay used in long transmission lines is
- a) Impedance relay b) Mho relay c) Reactance relay d) None of the above
- 36 In a 3 unit insulator string the voltage across lowest string is 10kV and the string efficiency is 90%, then the total string voltage will be
- a) 3kV b) 27kV c) 30kV d) 9kV
- 37 Which of the following circuit breakers are generally used in railway applications.
- a) Low oil circuit breaker b) Bulk oil circuit breaker c) SF₆ circuit breaker d) Air blast circuit breaker
- 38 In the design of a distributor, which of the following is the major consideration.
- a) Voltage drop b) Current capacity c) Frequency d) Mechanical strength
- 39 What is the type of insulator used for holding the HT overhead conductor on straight running of poles.
- a) Post insulator b) Pin type insulator c) Stay insulator d) Shackle insulator
- 40 Bundled conductors are mainly used in high voltage overhead transmission lines to
- a) Reduce transmission line losses b) Increase mechanical strength c) Reduce sag d) Increase in capacitance and reduce corona loss
- 41 Check linearity and time invariance of the given system, $y(t) = x(2t) + 3$
- a) Linear and time invariant b) Non-linear and time variant c) Non-linear and time invariant d) Linear and time variant
- 42 If P and E are the average power and energy of continuous or discrete time signal, then energy signal is
- a) $0 < E < \infty, P = 0$ b) $0 < E < \infty, P = \infty$ c) $0 < P < \infty, E = 0$ d) $0 < P < \infty, E = \infty$

- 43 What is the fundamental period 'T' of the signal, $x(t) = 2\cos(t/4)$
 a) 2π b) 4π c) 6π d) 8π
- 44 The Laplace transform of te^{at} is
 a) $1/s^2 - a$ b) $1/s^2 + a$ c) $1/(s-a)^2$ d) $s/s+a$
- 45 The final value of the system whose transfer function is represented by, $F(s) = (s+1)/s^2+2s+5$ is
 a) Unity b) 0.2 c) Infinity d) zero
- 46 The overall transfer function of the system given below is



- a) $G_1G_2/1+G_1H_1+G_1G_2H_1$ b) $G_1G_2/1+G_2H_2+G_1G_2H_1$ c) $G_1G_2/1+G_2H_2+G_1G_2H_2$ d) $G_1G_2/1-G_2H_2-G_1G_2H_1$
- 47 The characteristic equation of a system is given by, $P(s) = 2s^4+s^3+3s^2+5s+7=0$, the number of roots present in right half of s-plane is
 a) 3 b) 0 c) 1 d) 2
- 48 The capacitance in force-current analogy is analogous to
 a) Momentum b) Velocity c) Displacement d) Mass
- 49 The Z transform of $\delta(n-m)$ is
 a) Z^{-n} b) Z^{-m} c) $1/Z^{-n}$ d) $1/Z^{-m}$
- 50 The Z transform and ROC of the function, $f(n) = 3^n u(n)$ are
 a) $1/1-3Z^{-1}, |Z|>|3|$ b) $1/1-3Z^{-1}, |Z|<|3|$ c) $1/1+3Z^{-1}, |Z|>|3|$ d) $1/1+3Z^{-1}, |Z|<|3|$
